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COMPLIANCE MATRIX								
For								
INTELEGENT VEHICAL INTERCOM SYSTEM (iVIS)								
Sr. No.	Ref of Technical Specification and Quality Assurance Clause			Subject	Reference	Verification	Confirmation/ Deviation/ Exception/ Reservation	Remarks
	Section	Clause	Sub-Clause					
1	1			General. Digital Control Harness in AFVs and ICVs enable integration / full utilization of all facilities of the radio sets as also permit intercommunication with the crew.				
2	2			Physical Characteristics. The equipment should have the following physical characteristics				
3		(a)		Weight, Size and Volume				
4			(i)	The system should be light in weight and not impose any penalty on the functioning of the crew because of obstruction etc	Technical Manual Part I Vol 1 - Para 1.6.2 Technical Manual Part I Vol 1 - Para 1.7.2	ATP Para 8.2	Complied	
5			(ii)	The component units of the system should be capable of being fitted in the existing mounting points, with suitable adaptors, if required.		FFT ATP	Complied	
6		(b)		It should be durable and have an installed / shelf life of at least 15 years.			Complied	
7		(c)		The system should be protected against :-				
8			(i)	Reverse voltage. It should have circuit breakers / fuses of the appropriate current rating and other safety devices such as EMI filters, transient suppressors with suitable breakdown and clamping voltage.	Technical Manual Part I Vol 1 - Para 4.2.1	ATP Para 8.3.2 ATP Para 8.3.5	Complied	
9			(ii)	EMP to the extent possible.			Complied	
10			(iii)	Short / Open circuits.	Technical Manual Part I Vol 1 - Para 4.2.1	ATP Para 8.3.3	Complied	
11	3			Essential Features.				
12		(a)		It should be capable of providing intercommunication within the crew and integrating the following configuration of radio sets in armoured vehicles and their variants :-	Technical Manual Part I Vol 1 - Para 1.5.1 (3.)	ATP Para 8.9.1	Complied	
13			(i)	Command tank / ICV : 3 Sets (1x100 W HF set & 2x50 W VHF sets).	Technical Manual Part I Vol 1 - Para 1.6.1 (3.)	ATP Para 8.9.2	Complied	
14			(ii)	Control tank / ICV : 3 Sets (2x50 W VHF sets & 1x5 W VHF set).	Technical Manual Part I Vol 1 - Para 1.6.1 (3.)	ATP Para 8.9.3	Complied	
15			(iii)	Gun tank / ICV : 2 Sets (1x50 W VHF set & 1x5 W VHF set).	Technical Manual Part I Vol 1 - Para 1.6.1 (3.)	ATP Para 8.9.4	Complied	
16			(iv)	ARV: 1 Set (1x50W VHF set).	Technical Manual Part I Vol 1 - Para 1.6.1 (3.)	ATP Para 8.9.5	Complied	
17		(b)		It should be compact, simple and easy to operate.	Technical Manual Part I Vol 1 - Para 1.6.2 Technical Manual Part I Vol 1 - Para 1.7.2 Technical Manual Part I Vol 1 - Para 1.7.1 (3.)	ATP Para 8.2	Complied	
18		(c)		It should be rugged and be able to operate in an ambient temperature range from minus 20°C to plus 55°C and in all types of weather conditions	Technical Manual Part I Vol 1 - Para 1.2	Refer Environmental Test Report ULR-TC646824000005665F	Complied	
19		(d)		The system should occupy minimal space in the fighting compartment.	Technical Manual Part I Vol 1 - Para 1.6.2 Technical Manual Part I Vol 1 - Para 1.7.2	ATP Para 8.2	Complied	

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20	(e)	It should be easily integrated in existing and future armoured vehicles and the variants.	Technical Manual Part I Vol 1 - Para 1.5 (2.)	FFT ATP	Complied	
21	(f)	Its installation should be possible in the field, easy and not take more than 12 hours.		FFT ATP	Complied	
22	(g)	Installation should not require removal of the turret, or any major assemblies in the fighting compartment.		FFT ATP	Complied	
23	(h)	It should be fully solid state and digital in nature	Technical Manual Part IV Vol 1	Technical Manual Part IV Vol 1	Complied	
24	(i)	It should be capable to handling, routing and switching of min 04 voices and data channels.	Technical Manual Part I Vol 1 - Para 1.3 (2.)	ATP Para 8.9.1 ATP Para 8.9.3	Complied	
25	(k)	The voice channels must be digitized with at least 64 KB PCM	Technical Manual Part I Vol 1 - Para 1.3 (8.)	ATP Para 10	Complied	
26	(l)	The data channels must be capable of handling at least 48 KBPS data rate and the rate should be compatible with the digital radio sets being procured / installed.	Technical Manual Part I Vol 1 - Para 1.3 (9.)		Complied	Details required from end user regarding In Vehical Radio for Physical connector interface and software protocol.
27	(m)	It should support and enable simultaneous transmission of at least four VHF/HF radio sets of the latest digital technology	Technical Manual Part I Vol 1 - Para 1.3 (3.)	ATP Para 8.9.1	Complied	Supports In Service Radios
28	(n)	It should enable intercommunication within crew of the tks and BMP-II and also stick members of BMP-II, ie intercommunication between the commander, gunner, driver, section commander & sticks	Technical Manual Part I Vol 1 - Chapter 2	ATP Figure 17 ATP Figure 18	Complied	
29	(o)	The system should provide satisfactory wireless communication, with the crew members sitting in the armoured veh and stick members physically separated up to a distance of 1500m	Technical Manual Part I Vol 1 - Para 1.5.1 (4.) Technical Manual Part I Vol 1 - Para 1.9	ATP Para 8.9.4 FFT ATP	Complied	
30	(p)	It should be able to support a Battlefield Management System, which may be incorporated in the future.	Technical Manual Part I Vol 1 - Para 1.5 (10.)		Complied	Details required from end user regarding Battlefield Management System for Physical connector interface and software protocol.
31	(q)	It should be able to integrate other electronic systems in the BMP, like the navigation system (ALNS) / GPS Electronic voice warning systems, etc so that the crew can use the control harness to access them.	Technical Manual Part I Vol 1 - Para 1.3 (6.) Technical Manual Part I Vol 1 - Para 1.9	ATP Para 11 -	Complied Complied	Details required from end user regarding (ALNS)/GPS for Physical connector interface and software protocol. Details required from end user regarding Electronics Voice Warning System for Physical connector interface and software protocol.
32	(r)	It should be modular in nature and should be configurable to suit individual vehicle requirement (such as a tank, ICV etc), based on its task / role.	Technical Manual Part I Vol 1 - Para 1.5 (4.)	ATP Figure 17 ATP Figure 18	Complied	
33	(s)	The crew member units should have the facility to display short messages or commands, in the form of an alphanumeric character display.	Technical Manual Part I Vol 1 - Para 1.7.1 (4.)	ATP Para 8.7.2	Complied	
34	(t)	It should be programmable in field and support future upgrades.	Technical Manual Part I Vol 1 - Para 1.3 (5.)	UHB	Complied	
35	(u)	It should facilities repairs in situ by replacement of faulty modules.	Technical Manual Part II Technical Manual Part III	UHB	Complied	
36	4	Technical Characteristics.				
37	(a)	The system should have low power consumption.	Technical Manual Part I Vol 1 - Para 1.5 (6.)	ATP Para 8.3.1 ATP Para 8.3.4	Complied	
38	(b)	The system should have low power consumption. The system should draw power from the AFV batteries.	Technical Manual Part I Vol 1 - Para 1.5 (6.)	ATP Para 8.3.1 ATP Para 8.3.4	Complied	
39	(c)	The system should support both data and voice communication.	Technical Manual Part I Vol 1 - Para 1.3 (1.)	ATP Para 9 ATP Para 8.9.3	Complied	
40	(d)	There should be no interference when the radio sets and intercommunication system are being used concurrently.	Technical Manual Part I Vol 1 - Para 1.3 (4.)	FFT ATP	Complied	
41	(e)	The system should conform to Defence Standards and environmental conditions as laid down in JSS-55555	ESS Manual	Refer Enviourmental Test Report ULR-TC64682400005665F	Complied	

42		(f)	The system should meet MIL-STD-461E for EMI / EMC.	Technical Manual Part I Vol 1 - Para 4.2.1	Refer EMI/EMC Reports ULR No.: TC859725000000580F ULR No.:TC979725000000076F ULR No.: TC1200125000000205F	Complied	
43		(q)	The system components should be sealed against dust, humidity, fungi and insects during storage and usage conditions. The exposed parts should be resistant to corrosive actions of the atmosphere.	Technical Manual Part I Vol 1 - Para 1.5 (9.)	Refer Enviourmental Test Report ULR- TC646824000005665F	Complied	
45	5		MTBF. MTBF of 10,000 hours and MTTR of 06 hours is required.	Technical Manual Part I Vol 1 - Para 1.5 (8.)		Complied	
46	6		BITE. Built In Test facility should be provided and it should support features like loop tests and diagnostics and should enable a technician to trace a fault down to PCB level.	Technical Manual Part I Vol 1 - Para 2.2 (11.)	ATP Table 7	Complied	